

IMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 60%

QUESTIONS: 6

Total Marks:60

Annexure A

Application Based Questions

Code of Conduct:

*I (K. Deepak / 192320078) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Application Based Questions

1. Use Case Diagram Analysis

Question:

A Smart Hotel Reservation System allows guests to search rooms, make reservations, cancel bookings, and process payments. Managers handle room allocation and reports, while payment services process transactions.

Answer:

Analysis:

The system should clearly separate guest, manager, receptionist, and payment service responsibilities.

Actors:

- Guest
- Manager
- Receptionist
- Payment Service
- Admin

Use Cases:

- Register/Login
- Search Rooms
- Check Availability
- Make Reservation
- Cancel Booking
- Process Payment
- Generate Receipt
- Allocate Room
- Manage Rooms
- Generate Reports

Relationships:

- <<include>>: Make Reservation → Check Availability; Process Payment → Generate Receipt.
- <<extend>>: Cancel Booking extends Manage Reservation.
- Generalization: Manager and Receptionist inherit from Staff.

Text Diagram:

Guest → Search Rooms, Make Reservation, Cancel Booking

Manager → Allocate Room, Generate Reports

Receptionist → Manage Reservations

Payment Service ↔ Process Payment

2. Class Diagram Design

Answer:

Classes:

Student(studentId,name,email; enroll(),viewResult())

Faculty(facultyId,name; teachCourse(),evaluate())

Course(courseId,title,credits; assignFaculty())

Department(deptId,name; addCourse())

Examination(examId,date; conductExam(),publishResult())

Person(id,name)

Relationships:

- Inheritance: Student and Faculty inherit Person.
- Association: Student–Course, Faculty–Course, Course–Examination.
- Aggregation: Department contains Courses.

Multiplicity:

Department 1..* Courses

Course *.* Students

Faculty 1..* Courses

Course 1..* Examinations

Text Diagram:

Person

|-- Student

|-- Faculty

Department <|-- Course

Student -- Course

Faculty -- Course

Course -- Examination

3. Sequence Diagram Evaluation

Answer:

Objects:

Customer, Vehicle Rental App, Rental System, Payment Gateway, Vehicle Database.

Message Flow:

1. Customer → App: Search Vehicle
2. App → Database: Check Availability
3. Database → App: Available Vehicles
4. Customer → App: Book Vehicle
5. App → Rental System: Create Booking
6. Rental System → Payment Gateway: Request Payment
7. Payment Gateway → Rental System: Payment Success
8. Rental System → Customer: Rental Confirmation

Activation:

Each object becomes active while processing its request.

Lifelines continue until transaction completion.

4. State Diagram Construction

Answer:

States:

Initial, Red Signal, Green Signal, Yellow Signal, Emergency Mode, Maintenance Mode, Final.

Transitions:

Start → Red

Timer Expired → Green

Timer Expired → Yellow

Timer Expired → Red

Emergency Detected → Emergency Mode

Emergency Cleared → Red

Maintenance Request → Maintenance Mode

Maintenance Complete → Red

Shutdown → Final

5. Component Diagram Design

Answer:

Components:

- Patient Management
- Appointment Scheduling
- Medical Records
- Billing
- Notification Service
- Database

Interfaces:

Patient Management ↔ Appointment Scheduling

Appointment Scheduling ↔ Medical Records

Billing ↔ Database

Notification Service ↔ Patient Management

Dependencies:

Appointment depends on Patient Management.

Medical Records depend on Appointment.

Billing depends on Medical Records.

Notification depends on Appointment and Billing.

Architecture:

Users → Patient Management → Appointment Scheduling

Appointment → Medical Records → Billing

All Components → Database

Notification Service sends alerts.

6. Deployment Diagram Design

Answer:

Nodes:

- IoT Waste-bin Sensors
- Edge Device
- Cloud Server
- Database Server
- Monitoring Dashboard

Artifacts:

Sensors: Sensor Firmware

Edge Device: Data Collector

Cloud Server: Analytics Engine, Waste Management Application

Database Server: Waste Database

Dashboard: Web Monitoring Application

Communication:

Sensors → Edge Device (Wi-Fi/LoRa)

Edge Device → Cloud Server (Internet)

Cloud Server ↔ Database

Dashboard ↔ Cloud Server (HTTPS)

Text Deployment:

[IoT Sensors] -- [Edge Device] -- Internet -- [Cloud Server] -- [Database]

|
[Monitoring Dashboard]

RUBRICS FOR CALCULATION

Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand